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# A Review of RT-DETR-based Object Detection

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**Abstract**—The advent of Real-Time Detection Transformer (RT-DETR) represents a pivotal breakthrough in bridging the long-standing performance gap between real-time efficiency and high accuracy in object detection. This survey provides a comprehensive review of the RT-DETR framework, systematically analyzing its evolutionary trajectory from the original DETR model, its innovative hybrid CNN-Transformer architecture, and its benchmark performance that challenges the dominance of YOLO-series models. We first delineate the core components of RT-DETR—the CNN backbone, hybrid encoder, and Transformer decoder — and present a quantitative comparison with contemporary detectors, highlighting its exceptional speed-accuracy trade-off. Beyond model fundamentals, we consolidate and categorize a wide array of research efforts aimed at overcoming RT-DETR's limitations, focusing on four key improvement avenues: model lightweighting, feature fusion enhancement, training strategy optimization, and deployment efficiency. Furthermore, this review extensively documents the successful application of RT-DETR across diverse fields, including medical image analysis, military reconnaissance, industrial inspection, and agricultural automation, showcasing its adaptability and practical value. Finally, we discuss persistent challenges and outline promising future research directions, such as robustness in complex environments, multimodal fusion, and cross-domain generalization. This survey serves to contextualize RT-DETR's role in the evolving landscape of real-time object detection and to inspire continued innovation in this rapidly advancing field.

**Keywords**—RT-DETR, Object Detection, Real-time, Transformer, Applications, Future Directions.

## I. INTRODUCTION

Target detection, as a core task in computer vision, aims to simultaneously locate and classify objects of interest in images or video streams. It serves as the technical foundation for numerous downstream applications such as autonomous driving, intelligent surveillance, and industrial inspection. Over the past decade, the field has evolved through three key stages: traditional methods based on handcrafted features (e.g., HOG + SVM), two-stage detectors driven by deep learning (e.g., Faster R-CNN[1]), and one-stage detectors optimized for efficiency (e.g., YOLO, SSD[2][3][4]). Particularly, one-stage models have gained widespread adoption in real-world scenarios due to their balanced performance between speed and accuracy. However, with the increasing demand for real-time processing in edge devices and complex environments, the trade-off

between inference speed, detection precision, and robustness to challenging conditions (e.g., small targets, occlusions) remains a critical bottleneck.

Real-time target detection imposes stringent requirements on both algorithmic efficiency and detection quality, leading to several key challenges: Traditional one-stage models often sacrifice detection accuracy for faster inference, while two-stage detectors (e.g., Faster R-CNN) achieve higher precision but fail to meet real-time constraints (typically requiring >30 FPS for video streams). Small objects (e.g., distant pedestrians, tiny defects in industrial parts) suffer from insufficient feature representation due to limited pixel information, resulting in low recall rates. Variations in illumination (low-light, strong glare), background clutter, and object occlusions severely degrade detection performance, especially in dynamic scenarios such as UAV reconnaissance and autonomous driving. Edge devices have limited computational resources and power consumption budgets, requiring models to be lightweight and efficient.

RT-DETR (Real-Time DETR) is a groundbreaking real-time target detection model proposed by Baidu in 2023[5], which innovatively combines the advantages of DETR (Detection Transformer) with the efficiency of one-stage detectors. Unlike traditional DETR models that rely on pure Transformers and suffer from slow inference speed, RT-DETR introduces a hybrid encoder structure integrating CNN and Transformer, along with optimized attention mechanisms and lightweight design strategies. This enables it to achieve state-of-the-art performance in both speed and accuracy, breaking the long-standing speed-accuracy trade-off in real-time detection. It enriches the technical system of real-time target detection by verifying the feasibility of hybrid CNN-Transformer architectures, providing new ideas for solving the efficiency problem of Transformer-based detectors. With its excellent real-time performance and robustness, RT-DETR can be directly deployed in edge devices and complex scenarios, promoting the practical application of target detection in medical imaging, military reconnaissance, industrial inspection, and other fields. As a representative model of the next generation of real-time detectors, RT-DETR has sparked extensive research on lightweight Transformer design, multi-scale feature fusion, and edge deployment optimization, accelerating the technological iteration of the field.

## II. RT-DETR MODEL FUNDAMENTALS

### A. Brief History and Evolution from DETR

DETR (Detection Transformer), introduced by Facebook AI Research in 2020, revolutionized object detection with its end-to-end Transformer-based encoder-decoder architecture. By eliminating the need for handcrafted components such as anchor boxes and non-maximum suppression (NMS), it significantly streamlined the detection pipeline. Nevertheless, DETR faced two major limitations: (1) slower inference speed caused by the high computational complexity of the Transformer encoder, which scales quadratically ( $O(N^2)$ ) with the sequence length  $N$ ; and (2) difficulty in converging on small objects, necessitating extended training schedules — for instance, up to 500 epochs on the COCO dataset.

To address these issues, subsequent improvements such as Deformable DETR[6] and Conditional DETR[7] optimized the attention mechanism and training strategy, but their real-time

performance still fell short of one-stage models like YOLO. RT-DETR further evolves the DETR framework by:

1) Replacing the pure Transformer encoder with a hybrid encoder combining CNN and lightweight Transformer, reducing computational complexity.

2) Introducing a multi-scale feature fusion module tailored for real-time scenarios, enhancing small target detection capabilities.

3) Optimizing the training process with improved supervision mechanisms, shortening the convergence cycle.

This evolution makes RT-DETR the first DETR-based model that can compete with YOLO in real-time performance while maintaining high accuracy.

### B. Architecture Breakdown

RT-DETR adopts a modular architecture consisting of three core components: CNN Backbone, Hybrid Encoder, and Decoder with Attention Mechanism (Figure 1).

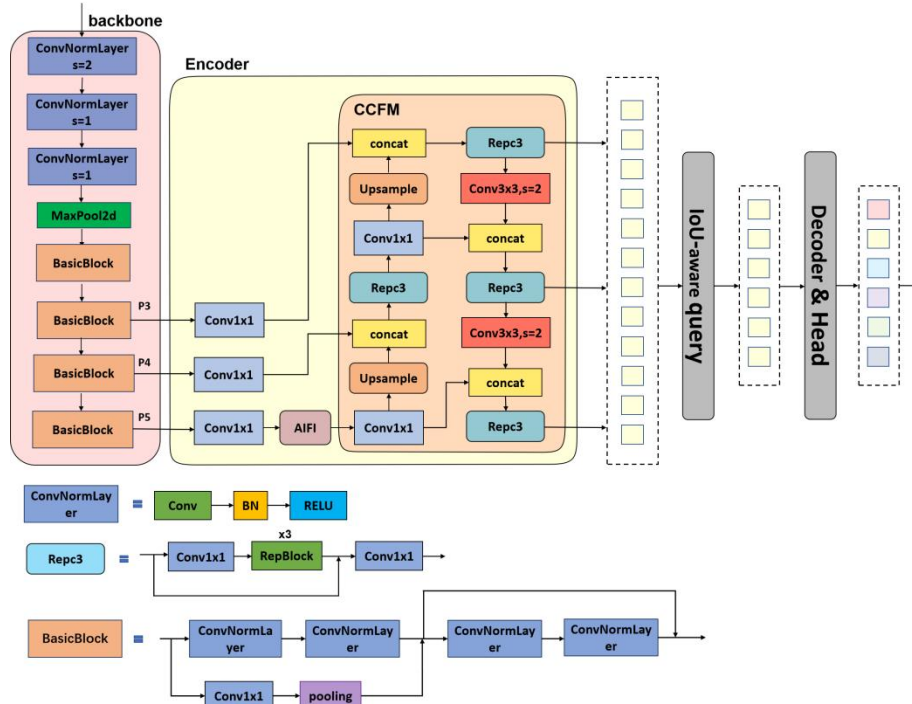


Fig.1. RT-DETR Network Structure

1) *CNN Backbone*: Mature architectures such as ResNet or CSPNet are adopted as the backbone, responsible for extracting discriminative low-level (e.g., edge, texture) and mid-level (e.g., local shape) features. Through a multi-stage downsampling process, the backbone generates three key scale feature maps: C3 (output of the 3rd stage, downsampled by  $8 \times$ ), C4 (output of the 4th stage, downsampled by  $16 \times$ ), and C5 (output of the 5th stage, downsampled by  $32 \times$ ). These multi-scale features provide hierarchical spatial information, laying a foundation for subsequent cross-scale fusion and small/large target detection. For CSPNet-based backbones, the Cross Stage Partial (CSP) connection is inherently

integrated — this design splits the feature map into two branches (one for direct propagation and the other for stacked convolution operations), which reduces computational cost by  $\sim 30\%$  while preserving feature richness, aligning with RT-DETR's real-time design goal.

2) *Hybrid Encoder*: As the core innovation of RT-DETR, the Hybrid Encoder integrates CNN-based local feature extraction and Transformer-based global information interaction, addressing the inefficiency of pure Transformer encoders. It consists of two cascaded sub-modules:

a) *Attention-based Intra-scale Feature Interaction (AIFI)*: For each input scale (C3, C4, C5), AIFI employs a

lightweight multi-head self-attention mechanism to model long-range dependencies within the same scale. Unlike standard Transformer attention ( $O(N^2)$  complexity), AIFI adopts a sparse attention design (e.g., window-based attention or strided sampling) to limit the interaction range, reducing computational overhead while capturing global context. This enhances the discriminability of single-scale features, especially for large targets requiring global semantic understanding. *The multi-head self-attention mechanism can be formulated as:*

$$\text{Attention}(Q, K, V) = \text{Softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (1)$$

Where  $Q, K, V$  represent query, key, and value matrices respectively, and  $d_k$  is the dimension of keys. In RT-DETR, this mechanism is optimized with sparse attention to reduce computational complexity from  $O(N^2)$  to approximately  $O(N\sqrt{N})$ .

*b) CNN-based Cross-scale Feature Fusion (CCFF):* This module fuses the intra-scale refined features across C3, C4, and C5 using depthwise separable convolutions and concatenation operations. It upsamples high-level features (C4, C5) and fuses them with low-level features (C3, C4) to supplement spatial details for small target detection, while downsampling low-level features to align with high-level semantic information. Compared to traditional FPN (Feature Pyramid Network), CCFF avoids complex top-down/bottom-up propagation, achieving efficient cross-scale fusion with linear computational complexity. The depthwise separable convolution operation consists of two parts:

$$\text{DepthwiseConv}(X) = X * K_{\text{depthwise}} \quad (2)$$

$$\text{PointwiseConv}(X) = X * K_{\text{pointwise}} \quad (3)$$

Where  $*$  denotes the convolution operation,  $K_{\text{depthwise}}$  is the depthwise convolution kernel applied per input channel, and  $K_{\text{pointwise}}$  is the  $1 \times 1$  convolution kernel that combines channel information. This design reduces computational cost significantly compared to standard convolution.

The Hybrid Encoder outputs a set of fused multi-scale feature sequences, which are fed into the decoder for target localization and classification.

*3) Decoder with Attention Mechanism:* The decoder adopts a Transformer decoder structure with learnable object

queries, responsible for converting the fused feature sequences into final detection results. Key characteristics include:

1) *Object Queries:* A fixed number of learnable query vectors are used to model potential target instances. These queries interact with the encoder's output features via cross-attention, adaptively extracting instance-specific information.

2) *Auxiliary Supervision Heads:* To accelerate convergence and mitigate the vanishing gradient problem, the decoder is equipped with auxiliary classification and regression heads at intermediate layers. During training, these heads provide supervision signals for each decoding step, enabling the model to learn hierarchical instance representations and reducing the required training epochs from 500 (in DETR) to  $\sim 100$ .

3) *End-to-End Prediction:* The final decoder layer outputs bounding box coordinates and class probabilities for each query. Unlike YOLO, RT-DETR does not require NMS post-processing, as the object queries inherently model distinct instances through attention mechanism, simplifying the inference pipeline.

This three-component architecture synergizes the efficiency of CNNs and the global modeling capability of Transformers, enabling RT-DETR to achieve a balance between real-time performance and detection accuracy.

### C. Comparison with Other Models

In the field of object detection, to comprehensively evaluate the advantages and characteristics of RT-DETR, it is necessary to conduct performance comparison analysis with mainstream object detection models. This section selects representative SOTA models, including single-stage detection algorithms (such as the YOLO series) and other Transformer-based detection models (such as DETR), using the COCO Val2017 dataset for experiments. The comparison focuses on key performance indicators such as detection accuracy (e.g., mAP value), inference speed (FPS), computational complexity (FLOPS), and small object detection capability. By constructing a performance comparison table, the table visually presents the performance of each model across different dimensions, clearly demonstrating RT-DETR's advantages and differences compared to other mainstream models in balancing real-time performance and accuracy, handling complex scenarios, and detecting small objects. This provides a solid basis for understanding its practical value and applicable scenarios.

TABLE I. COMPARISON OF REPRESENTATIVE OBJECT DETECTION ALGORITHMS ON THE COCO2017VAL DATASET

| Model        | BACKBONE | EPOCHS | PARAMS/M | GFLOPS | FPS <sub>bs=1</sub> | AP <sup>val</sup> | AP <sup>val</sup> <sub>50</sub> | AP <sup>val</sup> <sub>75</sub> | AP <sup>val</sup> <sub>S</sub> | AP <sup>val</sup> <sub>M</sub> | AP <sup>val</sup> <sub>L</sub> |
|--------------|----------|--------|----------|--------|---------------------|-------------------|---------------------------------|---------------------------------|--------------------------------|--------------------------------|--------------------------------|
| YOLOv5-L[8]  | -        | 300    | 46       | 109    | 54                  | 49.0              | 67.3                            | -                               | -                              | -                              | -                              |
| YOLOv5-X[8]  | -        | 300    | 86       | 205    | 43                  | 50.7              | 68.9                            | -                               | -                              | -                              | -                              |
| PPYOLOE-L[9] | -        | 300    | 52       | 110    | 94                  | 51.4              | 68.9                            | 55.6                            | 31.4                           | 55.3                           | 66.1                           |
| PPYOLOE-X[9] | -        | 300    | 98       | 206    | 60                  | 52.3              | 69.9                            | 56.5                            | 33.3                           | 56.3                           | 66.4                           |
| YOLOV6-L[10] | -        | 300    | 59       | 150    | 99                  | 52.8              | 70.3                            | 57.7                            | 34.4                           | 58.1                           | 70.1                           |
| YOLOV7-L[11] | -        | 300    | 36       | 104    | 55                  | 51.2              | 69.7                            | 55.5                            | 35.2                           | 55.7                           | 66.7                           |
| YOLOV7-X[11] | -        | 300    | 71       | 189    | 45                  | 52.9              | 71.1                            | 57.4                            | 36.9                           | 57.7                           | 68.6                           |

|                          |      |     |    |     |      |      |      |      |      |      |      |
|--------------------------|------|-----|----|-----|------|------|------|------|------|------|------|
| YOLOV8-L[12]             | -    | 300 | 43 | 165 | 71   | 52.9 | 69.8 | 57.5 | 35.3 | 58.3 | 69.8 |
| YOLOV8-X[12]             | -    | 300 | 68 | 257 | 50   | 53.9 | 71.0 | 58.7 | 35.7 | 59.3 | 70.7 |
| DETR[13]                 | R50  | 500 | 41 | 86  | 28   | 42.0 | 62.4 | 44.2 | 20.5 | 45.8 | 61.1 |
| DETR-DC5[13]             | R50  | 500 | 41 | 187 | -    | 43.3 | 63.1 | 45.9 | 22.5 | 47.3 | 61.1 |
| DETR-DC5[13]             | R101 | 500 | 60 | 253 | -    | 44.9 | 64.7 | 47.7 | 23.7 | 49.5 | 62.3 |
| Sparse DETR[14]          | R50  | 50  | 41 | 121 | 23.2 | 46.0 | 65.9 | 49.7 | 29.1 | 49.1 | 60.6 |
| Anchor-DETR-DC5[15]      | R50  | 50  | 39 | 172 | -    | 44.2 | 64.7 | 47.5 | 24.7 | 48.2 | 60.6 |
| Anchor-DETR-DC5[15]      | R101 | 50  | -  | -   | -    | 45.1 | 65.7 | 48.8 | 25.8 | 49.4 | 61.6 |
| Conditional-DETR-DC5[16] | R50  | 108 | 44 | 195 | -    | 45.1 | 65.4 | 48.5 | 25.3 | 49.0 | 62.2 |
| Conditional-DETR-DC5[16] | R101 | 108 | 63 | 262 | -    | 45.9 | 66.8 | 49.5 | 27.2 | 50.3 | 63.3 |
| Deformable-DETR[6]       | R50  | 50  | 40 | 173 | 19   | 43.8 | 62.6 | 47.7 | 26.4 | 47.1 | 58.0 |
| Efficient-DETR[17]       | R50  | 36  | 35 | 210 | -    | 45.1 | 63.1 | 49.1 | 28.3 | 48.4 | 59.0 |
| Efficient-DETR[17]       | R101 | 36  | 54 | 289 | -    | 45.7 | 64.1 | 49.5 | 28.2 | 49.1 | 60.2 |
| DINO-Deformable-DETR[18] | R50  | 36  | 47 | 279 | 5    | 50.9 | 69.0 | 55.3 | 34.6 | 54.1 | 64.6 |
| RT-DETR[5]               | R50  | 72  | 42 | 136 | 108  | 53.1 | 71.3 | 57.7 | 34.8 | 58.0 | 70.0 |
| RT-DETR[5]               | R101 | 72  | 76 | 259 | 74   | 54.3 | 72.7 | 58.6 | 36.0 | 58.8 | 72.1 |

To provide a clear and quantitative comparison of mainstream DETR variants, we evaluated several representative models on the COCO dataset using widely adopted performance and efficiency metrics. These metrics cover detection accuracy—including AP, AP50, AP75, and scale-specific APs/APm/API—as well as efficiency indicators such as parameter count, FLOPs, and inference speed (FPS). Table 1 summarizes the results[19].

Overall, the comparison reveals several notable observations. First, RT-DETR fundamentally breaks the long-standing trade-off in DETR-based models between accuracy and speed. Traditional DETR variants such as Sparse DETR can achieve decent accuracy but generally exhibit limited real-time capability, often below 20 FPS. In contrast, RT-DETR-R50 not only achieves 53.1 AP, significantly surpassing these models, but also delivers 108 FPS, marking a remarkable improvement in both dimensions simultaneously.

Second, RT-DETR demonstrates strong competitiveness against real-time CNN-based detectors. At similar accuracy levels, RT-DETR-R50 performs on par with state-of-the-art YOLO models like YOLOv8-L and YOLOv10-L, while maintaining high throughput. Conversely, among models operating at approximately 100 FPS, RT-DETR-R50 achieves substantially higher accuracy than earlier YOLO variants such as YOLOv5-L. This indicates that Transformer-based architectures, through the optimizations introduced in RT-DETR, are now capable of matching or even surpassing CNN-based detectors in real-time scenarios[20].

However, these advantages also come with certain limitations. Compared with pure CNN frameworks, RT-DETR's hybrid encoder structure is more complex and may require additional optimization for deployment. Moreover, its query selection mechanism, while improving efficiency, may risk missing objects in scenes with high density or numerous small targets.

Beyond RT-DETR, other efficiency-focused research can be broadly categorized into architecture simplification, model compression, lightweight network design, and hardware-aware optimization. These approaches collectively aim to further lower computation costs, enhance deployment flexibility, and adapt DETR-based models to diverse application scenarios[21].

### III. KEY IMPROVEMENTS AND INNOVATIONS

Despite the numerous advantages demonstrated by RT-DETR in the field of object detection, there are still some limitations:

- *Small Object Detection:* Although RT-DETR has shown improvements in small object detection compared to some traditional models, it may still miss extremely small targets (such as those with a very small pixel ratio and weak features).
- *Adaptability to Complex Environments:* Under extreme conditions, such as low light, strong noise, severe occlusion, and harsh weather, the detection performance of RT-DETR can fluctuate significantly. Its self-attention mechanism based on Transformer is susceptible to noise interference.
- *Model Lightweighting and Deployment:* For resource-constrained edge devices (such as low-power embedded chips and mobile devices), the computational load and parameter count of the RT-DETR model remain relatively high, making deployment challenging.
- *Multimodal Fusion:* RT-DETR lacks a mature and efficient fusion mechanism when integrating information from different modalities (such as RGB images with infrared images, depth data, etc.).

In response to these limitations of RT-DETR, researchers have conducted in-depth exploration and innovation in the following key areas:

#### A. Model Structure Optimization and Lightweighting

In the task of object detection, the computational complexity and inference speed of the model directly affect its ability for real-time applications. Traditional object detection models often struggle to operate efficiently on resource-constrained devices due to their large number of parameters and high computational overhead, which limits their deployment in scenarios such as mobile devices and embedded systems. To address these issues, researchers have optimized the model structure and introduced lightweight designs, which not only reduce the parameter count of RT-DETR but also significantly enhance its computational efficiency, making it more practical while maintaining high accuracy.

To meet the low-latency requirements for real-time detection, Li et al.[22] proposed LRT-DETR, which integrates the backbone network of YOLOv7-tiny, significantly reducing the number of parameters while improving detection speed. The SimpleUpsample operator was introduced in the encoder to reduce noise caused by upsampling, and a lightweight module called RepFormer was designed to effectively enhance feature extraction capabilities. The parameter count was compressed to 12M while maintaining high frame rate detection performance. This improvement makes it particularly suitable for real-time detection scenarios with limited computational resources, such as mobile devices or embedded systems.

Yu et al.[23] replaced the BasicBlock module with the Faster-CGLU module, while introducing the Hilo attention mechanism and BiFPN feature fusion strategy, optimizing the aggregation capabilities of local and global features and reducing computational complexity. This improvement significantly enhanced the model's efficiency and performance in real-time detection tasks.

#### B. Feature Fusion and Multi-Scale Modeling Optimization

In target detection tasks, relying solely on the features of the target itself often fails to achieve ideal detection results. Most existing detection methods overly focus on extracting local features of the target while neglecting the modeling of global semantic information and inter-target relationships in the scene, which directly reduces the model's ability to recognize small targets and adapt to complex scenarios. Zhang et al.[24] optimized the detection capability of multi-scale targets in remote sensing images by introducing Cascaded Group Attention (CGA) and Enhanced Bi-directional Feature Pyramid Network (EBiFPN), especially improving performance in small targets and complex backgrounds. This improvement enhanced the model's accuracy in processing multi-scale features. Zhou et al.[25] proposed a Multi-Scale Feature Sequence Fusion Real-Time Detection Transformer (MSRT-DETR), which integrates multi-scale feature sequences and utilizes network reparameterization techniques to achieve fast and accurate cell detection. By introducing a small target detection layer P2 and using the Scale Sequence Feature Fusion (SSFF) module to merge information from four different feature layers, the model enhanced detection accuracy

and robustness in high-density, multi-scale cell images by expanding the receptive field and optimizing the feature extraction process. This innovation significantly optimized the model's performance in dense target detection scenarios. Jin et al.[26] improved the backbone by combining ResNet-34 with Squeeze-and-Excitation Networks (SENetV2) and context-guided blocks to enhance shallow feature extraction and global information capture.

#### C. Training Strategy and Supervision Mechanism Optimization

In target detection tasks, traditional training strategies and supervision mechanisms often fail to fully leverage the potential of advanced models like RT-DETR. These methods typically rely on standard loss functions and label assignment techniques, lacking sufficient adaptability to complex detection tasks, especially in real-time scenarios that require rapid convergence and high-precision detection. This limitation may lead to inefficient training and restricted detection performance when facing diverse target appearances and scales. To address these issues, researchers have proposed a series of innovative training strategies and supervision mechanisms for RT-DETR to enhance the model's convergence speed and detection accuracy.

Lv et al.[27] proposed a Bag-of-Freebies-based training optimization scheme in RT-DETRv2. This method integrates a series of optimization techniques that do not increase inference latency, such as data augmentation and label smoothing. Data augmentation helps the model generalize better to different scenarios by generating diverse training samples, while label smoothing reduces the risk of overfitting by softening label distributions, further enhancing detection performance. The label smoothing technique can be formulated as:

$$y_{smooth} = (1 - \epsilon) \cdot y_{one-hot} + \frac{\epsilon}{K} \quad (4)$$

Where  $\epsilon$  is the smoothing factor (typically 0.1),  $y_{one-hot}$  is the original one-hot encoded label, and  $K$  is the number of classes. This regularization technique prevents the model from becoming overconfident during training.

These strategies significantly optimized RT-DETR's performance in real-time detection tasks, especially in complex backgrounds and multi-target scenarios, without increasing computational costs. Wang et al.[28] introduced an innovative training method in RT-DETRv3 by adding a CNN auxiliary branch to provide dense supervision, significantly enhancing the model's capability in multi-scale feature learning. This method compensates for the shortcomings of traditional sparse supervision in complex scenarios by providing finer-grained supervision signals for each feature layer. Additionally, they designed a self-attention perturbation strategy to enrich the label assignment of positive samples. This strategy enhances the diversity of positive samples by introducing perturbations, allowing the model to better adapt to variations in target appearance and position, thereby improving its robustness. These improvements not only accelerated the model's convergence speed but also significantly increased detection accuracy, especially in applications with high real-time

requirements. The overall loss function in RT-DETR combines multiple components:

$$L = \lambda_{cls} \cdot L_{cls} + \lambda_{box} \cdot L_{box} + \lambda_{giou} \cdot L_{giou} \quad (5)$$

where  $L_{cls}$  is the classification loss (typically focal loss),  $L_{box}$  is the bounding box regression loss (L1 loss), and  $L_{giou}$  is the Generalized IoU loss for better box overlap measurement. The  $\lambda$  coefficients balance the contributions of each loss term.

#### D. Model Deployment and Energy Efficiency Optimization

In practical applications, deployment efficiency and energy performance are core factors determining the usability of a model. Although RT-DETR has significantly reduced model complexity through a series of optimizations, its real-time performance still needs improvement in actual deployment scenarios. To address this pain point, researchers have proposed various targeted optimization strategies aimed at further enhancing RT-DETR's deployment adaptability in resource-constrained environments.

Liu et al. [29] proposed a fully integer-based quantization inference scheme, which effectively serves as a data compression method to reduce memory usage and computational complexity. Unlike previous methods that only quantized the linear layers of DETR, this approach further applies integer approximation to all nonlinear operation layers (e.g., sigmoid, Softmax, LayerNorm, GELU), enabling the entire inference process to be executed in the integer domain. The quantization process can be expressed as:

$$X_{int} = \text{round}\left(\frac{X_{float} - Z}{S}\right) \quad (6)$$

where  $S$  is the scale factor,  $Z$  is the zero point,  $X_{float}$  is the original floating-point value, and  $X_{int}$  is the quantized integer value. This transformation allows the model to run efficiently on integer-only hardware.

Yang et al. [30] proposed a knowledge framework combining feature-logic distillation, using a pruned optimized model as the student model and RT-DETR-R50 as the teacher model, effectively addressing the information loss caused by structural differences between the teacher and student, ensuring good detection performance while achieving model lightweighting.

#### IV. RT-DETR APPLICATION EXAMPLES

The RT-DETR algorithm, with its core innovations such as "CNN+Transformer hybrid architecture," "deformable attention mechanism," and "dynamic label matching," has achieved breakthroughs in balancing real-time performance and detection accuracy. Its global context modeling capability and advantages in multi-scale feature processing demonstrate unique value in complex scenarios across various fields, including healthcare, military, industrial inspection, transportation, and agriculture. Specific applications are as follows:

##### A. Medical Image Analysis

In medical imaging, early lesions (such as pulmonary nodules and retinal microaneurysms) are often characterized by

small size and vague features, making them easily confused with normal tissue textures. Traditional CNN models, which rely on local feature extraction, have limited capability in detecting such lesions. However, RT-DETR, with its global modeling ability and advantages in small target detection, has become a key solution to this problem.

To achieve rapid and accurate detection of ground-glass nodules while reducing interference from surrounding tissues such as bones and muscles, Tang et al. [31] proposed a targeted improvement plan: first, a method combining threshold segmentation and morphology was used for lung parenchyma segmentation, accurately separating lung tissue from surrounding non-target structures; second, the Transformer self-attention mechanism of RT-DETR was utilized to effectively associate the spatial relationship between "nodules and lung lobes" (for example, the clinical feature that small nodules close to the pleura have a higher probability of malignancy), enhancing the accuracy of lesion localization. Additionally, a ConvGLU module was designed to replace two  $1 \times 1$  convolution layers in FasterNet, preserving the edge details of 1-3mm small nodules through multi-scale feature fusion, and integrating the ConvGLU module into the FCGE module, significantly improving the detection capability for edge-blurred ground-glass nodules. The specific structure of the ConvGLU module is shown in Figure 2 [32] [33].

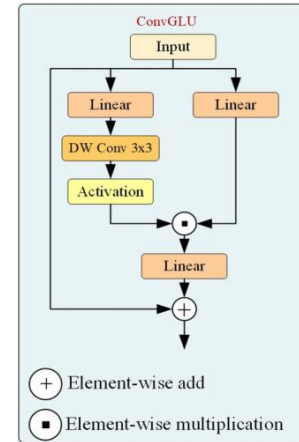


Fig. 2. ConvGLU Module

ConvGLU, as a hybrid mechanism, combines Gated Linear Units (GLU) [34] with Depthwise Convolution (DWConv). Its core function is to enhance information interaction between channels: its gating mechanism allows the model to dynamically adjust channel responses, enabling it to focus on key features even when faced with edge-blurred lesions; while depthwise convolution can accurately capture detailed information in local areas (including edge features), which is crucial for identifying irregularly shaped and poorly defined ground-glass nodules. The specific structure of the FCGE module is shown in Figure 3 [35].

Research shows that the lung nodule detection system based on RT-DETR achieved a sensitivity (true positive rate) of 92.3% on the LIDC-IDRI dataset, an improvement of 8.5% over YOLOv8, particularly with a 12% increase in detection

rate for ground-glass nodules (the type with the most vague features) [31].

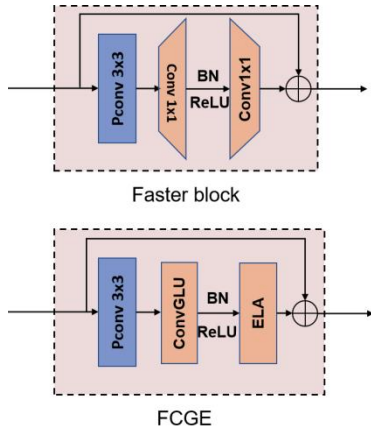


Fig.3. FCGE Module and Faster Block Module

In the diagnosis of diabetic retinopathy, RT-DETR can accurately identify microaneurysms (small targets) and exudates in fundus images that are less than  $20\mu\text{m}$ , using global features such as "lesion distribution density" to assist in grading. Its mean average precision (mAP@0.5) on the DDR dataset reached 89.7%, reducing the missed detection rate by 15% compared to traditional Faster R-CNN, thus gaining time for early intervention.

### B. Military and Defense Field

The military scenario has stringent requirements for "long-range small target detection," "interference resistance in complex backgrounds," and "real-time response," such as in drone reconnaissance and remote sensing image detection tasks. The hybrid architecture of RT-DETR can simultaneously meet the demands for "long-range feature preservation" and "real-time decision-making."

1) *Drone Aerial Detection*: In low-altitude reconnaissance missions with drones, ground vehicles often appear as small targets of less than  $32 \times 32$  pixels in images and are easily obscured by vegetation and buildings.

Zhao et al.[36] increased the detection layers for small targets and proposed a multi-scale feature fusion module to enhance the perception capability for small targets. They optimized the backbone network by replacing the traditional Basic Block with the MCP Block, making the model more lightweight while improving its feature extraction ability for small targets. Wei et al.[37] proposed integrating recalibration attention units and deformable attention mechanisms in the neck network to strengthen multi-scale feature fusion and improve localization accuracy; they introduced the Focaler-IoU loss function to better address the issue of uneven target scale distribution while focusing on hard sample learning.

2) *Remote Sensing Object Detection*: In remote sensing detection tasks, due to the large variation in target scales, the predominance of small targets, and complex backgrounds in remote sensing images, remote sensing object detection still faces challenges in practical applications, such as low

detection accuracy, poor real-time performance, high missed detection rates, and high false detection rates[38].

Zhang et al.[39] proposed a new remote sensing detection transformer (RS-DETR) model. First, the CGA was introduced into the AIFI module to enhance intra-scale feature interaction and improve computational efficiency. To address the issues of complex backgrounds and numerous small targets in remote sensing images, an EIFPN method was proposed to enhance cross-scale feature fusion. By combining the rich low-level features of small target information, the accuracy of small target detection was improved. Additionally, the Focaler-GIoU loss function was used as the localization regression loss, reducing sensitivity to scale variations and improving overall localization accuracy.

### C. Industrial Inspection Field

1) *Railway Turnout Defect Detection*: Railway turnouts are critical components of the rail track system, and their defects can lead to severe safety incidents and significant property damage. The irregular distribution and varying sizes of railway-turnout defects, combined with changing environmental lighting and complex backgrounds, pose challenges for traditional detection methods, often resulting in low accuracy and poor real-time performance.

Yu et al.[40] proposed a high-precision recognition model FHB-DETR based on the RT-DETR architecture, replacing the convolution in the original BasicBlock module with Faster-CGLU, significantly reducing the model's parameter count and computational load. The introduction of Hilo attention improved inference speed and FPS, while the feature extraction part incorporated bifpn, further enhancing feature extraction efficiency.

2) *Bearing Defect Detection*: Bearings are indispensable components in chemical equipment, serving as key supports for rotating shafts. Their performance profoundly affects the stability and reliability of the equipment. Insufficient bearing functionality can lead to imbalances in rotating parts, resulting in increased vibrations and noise, and in severe cases, may even cause equipment failure or damage[41].

Liu et al. [42] chose to use RT-DETR as the baseline model and proposed the Bearing-DETR model. First, by integrating Dysample dynamic upsampling, the model's feature processing capability is enhanced, allowing for more detailed and accurate defect identification. Second, an efficient model optimization (EMO) with Meta-Mobile Blocks (MMB) is introduced to optimize the model architecture for better performance and faster convergence. Finally, a Deformable Large Kernel Attention (D-LKA) is added, significantly improving the model's ability to adapt to different defect shapes and sizes, effectively enhancing detection accuracy[43].

#### D. Agricultural Detection Field

In agricultural scenarios, early features of crop pests and diseases (such as aphids and leaf spot disease) are small (e.g., a single aphid <1mm), sparsely distributed, and easily confused with leaf textures. Meanwhile, the backgrounds in drone aerial images, such as crop rows and field ridges, are complex, making traditional models prone to misjudgment. The small target detection and scene semantic modeling capabilities of RT-DETR can accurately address these issues[44][45].

##### 1) Ripe Tomato Detection:

In the detection of ripe tomatoes, Wang et al.[46] proposed a fine-grained recognition method called PDSI-RTDETR for assessing tomato maturity, providing more accurate decision support for intelligent grading and harvesting management. By integrating partial convolutional blocks with residual blocks in the PConv\_Block module, the traditional Basic\_Block structure in the backbone was replaced, reducing computational load and total parameters. The combination of deformable attention module and AIFI module enhanced the model's ability to extract detailed tomato features, improving both accuracy and efficiency. Additionally, a Slimneck-SSFF feature fusion architecture was proposed as a feature fusion framework, with optimized loss functions and the introduction of Inner-EIoU to accelerate convergence. Auxiliary frames were also utilized to enhance small object detection capabilities. When applied to future intelligent harvesting robots, this improved method can enhance tomato harvest quality by reducing the collection of immature and spoiled fruits.

##### 2) Cotton field insect detection:

To address the frequent issues of low accuracy, missed detections, and false positives in cotton field insect detection, Chen et al.[47] proposed an improved RT-DETR-based algorithm. The method first replaces the second traditional convolution in residual blocks with WTConv, significantly expanding the receptive field while maintaining fewer trainable parameters, enabling better focus on small target detection. It then introduces an M2SA module with a dual-branch architecture to extract global features and channel information, enhancing the model's understanding of complex scenes and improving small target detection accuracy. Finally, a small target optimization pyramid is proposed during cross-scale feature fusion, effectively learning both global and local features to optimize detection performance. This provides an efficient identification method for precise management and control of cotton field pests.

#### V. FUTURE DEVELOPMENT DIRECTIONS

Although RT-DETR is currently widely used in various industries and demonstrates outstanding performance, it still faces numerous challenges and opportunities for optimization. The model has room for improvement in handling non-uniform data distributions, high-complexity visual backgrounds, and cross-domain generalization capabilities. Future research can be conducted in the following areas:

- Further enhance the model's multi-scale feature learning ability to improve the representation and recognition accuracy of targets at different scales in complex scenes

- Enrich the quality and diversity of the training dataset to build a high-quality annotated sample library covering multiple scenes and modalities;
- Integrate multi-modal data sources, combining images, text, speech, and structured information to achieve a detection system with enhanced semantic understanding capabilities.

#### A. Medical Image Analysis Field

In the medical field, RT-DETR has not only achieved good applications in diabetic retinopathy detection but can also be further expanded to other automated image analysis tasks, such as macular degeneration, glaucoma, and brain lesion detection. Future research can focus on the following directions:

1) Strengthen the multi-scale feature modeling capability to accurately capture multi-level features from small lesions to large lesions, achieving intelligent diagnosis across different diseases;

2) Improve the quality and coverage of the dataset by including medical images of various diseases, different imaging stages, and under multiple device conditions, ensuring annotation accuracy and diversity, thereby enhancing the model's generalization ability;

3) Integrate multi-modal medical data, combining patient history, physiological measurements, and imaging features to construct a unified diagnostic model for more accurate disease prediction and classification diagnosis.

Through the development of the above directions, RT-DETR is expected to play a greater role in medical auxiliary diagnosis, promoting the evolution of intelligent medical image analysis towards precision and comprehensiveness.

#### B. Military Reconnaissance and Defense Applications

In the military and defense sector, RT-DETR still faces the issue of insufficient detection robustness in complex environments. Extreme lighting conditions (such as overexposure, low light, fog, rain, etc.) and high dynamic backgrounds often lead to a decline in detection performance. Future research can focus on the following aspects:

1) Optimize data augmentation and image preprocessing strategies, designing adaptive enhancement methods for different extreme environments to improve the model's robustness under harsh conditions;

2) Explore model lightweighting and efficient inference mechanisms to reduce parameter scale and computational load while ensuring accuracy, enabling deployment in resource-constrained drones, edge terminals, and weapon guidance systems;

3) Research environment-adaptive detection frameworks to achieve dynamic transfer and online tuning of models in multi-domain tasks.

Through these improvements, RT-DETR can provide more reliable technical support in key tasks such as battlefield target recognition, weapon guidance, and multi-source reconnaissance[21].

### C. Industrial Inspection and Quality Control

Although RT-DETR shows strong advantages in industrial defect detection, it is still limited by the high noise characteristics of industrial data and the issue of sample imbalance. Future research can focus on the following directions:

- 1) Construct adaptive multi-scale feature fusion networks that self-adjust feature weights for defects of different sizes and shapes, enhancing the model's ability to capture small defects;
- 2) Expand and standardize industrial datasets to cover more product types and environmental scenarios, improving the model's cross-domain generalization performance;
- 3) Introduce multi-modal detection mechanisms that integrate thermal imaging, depth maps, and spectral information to enhance detection accuracy;
- 4) Optimize model efficiency and deployment performance by utilizing model quantization, pruning, and hardware acceleration techniques to achieve real-time defect detection and dynamic feedback on high-speed production lines.

These improvements will further enhance the usability and stability of RT-DETR in smart manufacturing and industrial quality inspection scenarios.

### D. Smart Agriculture and Ecological Monitoring

In the agricultural sector, RT-DETR has been applied to pest detection and crop maturity assessment, but its adaptability in complex environments and diverse planting patterns still needs improvement. Future research directions include:

- 1) Strengthening the model's dynamic feature learning capability to address the diversity of crop morphology and environmental changes through deformable convolution and adaptive attention mechanisms;
- 2) Improving the quality and diversity of agricultural datasets by collecting image samples from different regions, crop types, and growth stages, and refining the annotation system;
- 3) Integrating multimodal agricultural information by combining multispectral, LiDAR, and meteorological data to achieve more accurate pest identification and yield prediction;
- 4) Simplifying model deployment and usage by developing lightweight versions to adapt to drones, field terminals, and mobile devices, promoting the popularization and sustainable development of smart agriculture[48].

## VI. CONCLUSION

In summary, RT-DETR, as an important innovative achievement in the field of object detection, has made significant breakthroughs in real-time performance and accuracy due to its unique architectural design, providing powerful solutions for complex scene detection tasks across multiple industries. From aiding early disease diagnosis in medical image analysis to addressing target monitoring in harsh environments for military reconnaissance, and ensuring

production safety in industrial inspections as well as supporting precision production management in agriculture, its application range is broad and its value is significant. However, as application scenarios continue to expand and deepen, RT-DETR also faces numerous challenges, such as dealing with uneven data distribution, interference from complex visual backgrounds, and higher demands for accuracy, efficiency, and lightweight solutions across different fields. Fortunately, many scholars are actively exploring its limitations, continuously innovating in model structure optimization, feature fusion, training strategies, and specific applications in various fields, thereby enhancing the performance and applicability of RT-DETR.

Looking ahead, under the trend of multidisciplinary technological integration, RT-DETR is expected to better overcome existing limitations and unlock more potential application scenarios by further enhancing multi-scale feature learning, enriching datasets, integrating multimodal data, and optimizing computational efficiency. This will continue to promote the deep application of object detection technology across various industries, contributing greater strength to the realization of intelligent and automated societal development goals. Its subsequent development is worth our ongoing attention and anticipation.

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